

Joaquin Gonzalez-Salgado

jgonzalezsalgado@hmc.edu | [jgonzalezsalgado.github.io](https://github.com/jgonzalezsalgado) | [linkedin.com/in/joaquin-e-gs](https://www.linkedin.com/in/joaquin-e-gs)

EDUCATION

Harvey Mudd College

B.S. in Engineering (Electrical & Computer Engineering)

Claremont, CA

Expected May 2028

EXPERIENCE

Electrical Engineering Clinic

Aug. 2026 – Present

Blue Origin

CA

- Lead the electrical sub-team developing a full-scale sensor package for autonomous mating of ground- and vehicle-side launch umbilical panels carrying propellant, pneumatic, hydraulic, data, and power connections
- Conduct systems trades across candidate sensing technologies against operational requirements, weighing cost, availability, reliability in harsh launch environments, and EMI risk to vehicle avionics
- Own component selection, sensor installation and packaging design, and hardware procurement; work cross-functionally with Blue Origin engineers on prototype build-up and testing of a full-scale working model

Hardware Researcher

Aug. 2025 – Present

USDA — Electropenetrography (EPG) Amplifiers

CA

- Design, bring up, and test low-noise, field-deployable mixed-signal circuits: analog front-end amplifiers feeding microcontroller data acquisition for monitoring disease-vector insects
- Debug and characterize prototypes with oscilloscopes, function generators, and power supplies, iterating to improve signal quality and reliability for field deployment
- Develop microcontroller firmware and Python software to acquire, process, and visualize high-sensitivity analog signals

Physics Researcher

Aug. 2024 – Aug. 2025

Drone FPGA Radio-Telescope Beam Mapping

CA

- Built an FPGA-based software-defined radio in Verilog and Python, transmitting and receiving GHz-wideband signals for FFT-based DSP analysis
- Integrated a custom Xilinx FPGA RF transmitter onto a hexacopter drone to calibrate and map a ground-based radio telescope's beam pattern
- Integrated and debugged a multi-subsystem platform spanning flight control (PX4/ArduPilot), GPS synchronization, RF amplification, and power, taking it from bench to field deployment

PROJECTS

EV Design Lab | 36V Battery System, Power Distribution, Circuit Protection

Jan. 2026 – Present

- Lead a team build of an electric vehicle, integrating a 36V battery pack, motor controllers, relays, fusing, and circuit protection
- Designed the electrical system architecture and power distribution for safety, energy management, and serviceability; debugged and validated it through integration with the drivetrain
- Fabricated the chassis and mounted the battery enclosure and electronics for weight, stability, and serviceability

Autonomous Underwater Vehicle | Teensy 4.0, Sensors & Actuators, Kp Control

Jan. 2026 – Present

- Designed a custom motherboard around a Teensy 4.0 with op-amp signal conditioning for temperature and depth sensors, then brought up and debugged the board
- Wrote firmware implementing proportional (Kp) control to drive the vehicle's actuators from GPS position
- Built the vehicle for about \$50 from low-cost materials and tuned it to neutral buoyancy for safe deployment

TECHNICAL SKILLS

Electrical Design: Circuit Design, Schematic Capture, PCB Layout, Board Bring-Up & Debug, Mixed-Signal / Analog Front Ends, Power Distribution, Systems Trades, Component Selection

ECAD & Simulation: Altium Designer, KiCad, LTspice, LabVIEW, MATLAB

Firmware & Embedded: Microcontrollers (Teensy), FPGAs (Xilinx, Lattice), I2C, SPI, UART, CAN bus, Bluetooth

Languages: C, C++, Python, Verilog, VHDL, MATLAB

Lab & Tools: Oscilloscopes, Function Generators, Power Supplies, Multimeters, Git, CMake, Quartus, Questa, Fusion CAD